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## COMPUTER-READABLE RECORDING MEDIUM, ABNORMALITY DETECTION METHOD, AND INFORMATION PROCESSING DEVICE


### Abstract

A non-transitory computer-readable recording medium stores therein an abnormality detection program that causes a computer to execute a process including calculating a first value and a second value, based on a detection result of an object detected from an input scene to an object detection model and a saliency map for the input scene obtained by an XAI technique, the first value indicating a value of the saliency map for the entire input scene, the second value indicating a value of the saliency map in a region other than a region of the detected object in the input scene, and detecting an abnormality in the input scene based on the calculated first and second values.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and incorporates by reference the entire contents of Israel Patent Application No. 311286 filed on Mar. 6, 2024.

Field

[0002] The embodiments discussed herein are related to a computer-readable recording medium, an abnormality detection method, and an information processing device.

BACKGROUND

[0003] Object detection models (object detector: OD) are known, which in response to input of image data, output the region of an object in the image data and information (for example, label) of the object in the image data. Object detection models are used in various fields, including the retail field and the field of suspicious person detection, to detect various objects in response to input of an input scene which is an example of image data. Examples of object detection models used include those that identify products picked up or purchased by users at retail stores, and those that detect suspicious objects at airports.

[0004] The input scene to such an object detection model may contain an abnormality. This “abnormality” may be caused by an attacker putting an adversarial patch deliberately, or it may occur naturally by object hiding another object or an unrecognized object.

[0005] Detecting deliberate patch attacks, can be done for example by using training data containing an adversarial patch to perform training such that a scene containing an adversarial patch is classified into a new class (adversarial class) indicating that an attack has occurred, thereby detecting the presence/absence of an adversarial patch.

[0006] Non Patent Document 1: Ji, N., Feng, Y., Xie, H., Xiang, X., and Liu, *“Adversarial YOLO: Defense Human Detection Patch Attacks via Detecting Adversarial Patches”* 2021, arXiv preprint arXiv: 2103.08860.

[0007] However, in the current situation where specialized algorithms are designed for each individual type of abnormalities, it is difficult to detect an abnormality other than the targeted types. On the other hand, in practical applications in various fields such as retail and suspicious person detection, the type of abnormality contained in an input scene to an object detection model is often unknown. The above technologies therefore have room for improvement in terms of detecting any type of abnormality.

SUMMARY

[0008] According to an aspect of an embodiment, a non-transitory computer-readable recording medium stores therein an abnormality detection program that causes a computer to execute a process including calculating a first value and a second value, based on a detection result of an object detected from an input scene to an object detection model and a saliency map for the input scene obtained by an XAI technique, the first value indicating a value of the saliency map for the entire input scene, the second value indicating a value of the saliency map in a region other than a region of the detected object in the input scene, and detecting an abnormality in the input scene based on the calculated first and second values.

Advantageous Effects of Invention

[0009] The object and advantages of the invention will be realized and attained by means of the elements and combinations particularly pointed out in the claims.
[0010] It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory and are not restrictive of the invention.

## Description

BRIEF DESCRIPTION OF DRAWINGS
[0011] FIG. **1** is a diagram illustrating a system configuration example;
[0012] FIG. **2** is a schematic diagram illustrating one aspect of a problem-solving approach;
[0013] FIG. **3** is a block diagram illustrating a functional configuration example of a server device;
[0014] FIG. **4** is a schematic diagram illustrating an example of object detection;
[0015] FIG. **5** is a schematic diagram illustrating an example of saliency map generation;
[0016] FIG. **6** is a schematic diagram (**1**) illustrating an example of DiL score calculation;
[0017] FIG. **7** is a schematic diagram (**2**) illustrating an example of DiL score calculation;
[0018] FIG. **8** is a schematic diagram (**3**) illustrating an example of DiL score calculation;
[0019] FIG. **9** is a schematic diagram (**4**) illustrating an example of DiL score calculation;
[0020] FIG. **10** is a flowchart illustrating an abnormality detection process; and
[0021] FIG. **11** is a diagram illustrating a hardware configuration example.
DESCRIPTION OF EMBODIMENTS
[0022] Preferred embodiments will be explained with reference to accompanying drawings. The embodiments only describe one example or aspect by way of illustration, and such illustration is not intended to limit structures, actions, functions, properties, characteristics, methods, applications, and the like according to the present disclosure.
First Embodiment
Overall Configuration
[0023] FIG. **1** is a diagram illustrating a system configuration example. FIG. **1** illustrates, as an exemplary use case, an example in which an abnormality detection function of detecting an abnormality contained in an input scene to an object detection model is applied to an automatic checkout system **1** for stores such as supermarkets and convenience stores. As used herein “automatic checkout system” may be referred to as “self-checkout system”.
[0024] In the following, an example in which the above abnormality detection function is packaged and provided with the automatic checkout service will be described by way of illustration. The above abnormality detection function may be provided independently of the above automatic checkout service. In this case, the automatic checkout service and the abnormality detection function may be provided by different hardware resources, for example, physical devices and physical elements.
[0025] As illustrated in FIG. **1**, the automatic checkout system **1** can include a server device **10**, a camera **20**, and a self-checkout register **30**. Although FIG. **1** illustrates the camera **20** and the self-checkout register **30** as components of the automatic checkout system **1**, other devices, such as terminal devices used by store personnel, may also be included.
[0026] These server device **10**, camera **20**, and self-checkout register **30** may be connected via a network NW. For example, the network NW may be either wired or wireless and implemented by any technology, such as an intranet, the Internet, or a power-saving wireless communication standard for the Internet of things (IoT). The network NW is not necessarily a single network and may include multiple networks in which an intranet and the Internet are connected via a gateway or the like.
[0027] The server device **10** is an example of an information processing device that provides the above abnormality detection function. For example, the server device **10** can be implemented as an

application that is software as a service (Saas) based to provide the above abnormality detection function as a cloud service. Alternatively, the server device **10** can be implemented as a web server that provides the above abnormality detection function on the premises.
[0028] The camera **20** is an example of an imaging device that captures videos. For example, the camera **20** is installed to be able to take a picture of a specific range within a store **3**, for example, an area that includes the self-checkout register **30** in the checkout corner. With this configuration, for example, a composition in which a user at the store pays at the self-checkout register **30** can be taken.
[0029] The video data thus captured by the camera **20** may be transmitted to the server device **10**. For example, the video data may include a plurality of image frames captured by the camera **20** in chronological order. A frame number corresponding to the ascending chronological order is added to each of the image frames. Such an image frame corresponds to each of the still images (frames) included in moving images captured by the camera **20** and can be equivalent to an example of an input scene to an object detection model described above.
[0030] The self-checkout register **30** is an example of an accounting machine with which users who purchase products register and settle (pay) for the purchased products by themselves. The self-checkout register **30** is called "self checkout", "automated checkout", "self-checkout machine", "self-checkout register", or the like.
One Aspect of Problem
[0031] As explained in the section of BACKGROUND, in practical applications in various fields such as retail and suspicious person detection, the type of abnormality contained in an input scene to an object detection model is often unknown.
[0032] In one aspect, "abnormality" may occur naturally due to "partial occlusion" or "out-of-distribution". As used herein "partial occlusion" may refer to a phenomenon in which an object in the foreground of a camera subject partially hides an object behind. The term "out-of-distribution" may refer to a phenomenon in which an object serving as a subject belongs to a class of domains that falls outside the distribution of a training dataset of an object detection model. In another aspect, "abnormality" may be caused deliberately by an attacker putting an adversarial patch in a physical or digital space.
[0033] In an aspect, system robustness can be acquired only after any type of abnormality, including these types such as partial occlusion, out-of-distribution, and adversarial patches, can be detected.
[0034] However, in the current situation where specialized algorithms are designed for each individual type of targeted abnormalities, it is difficult to detect an abnormality other than the targeted types. For example, an algorithm specialized for detecting patch attacks fails to evaluate the effect of a naturally occurring abnormality on the decision-making process of object detection models. On the other hand, an algorithm specialized for partial occlusion fails to evaluate the effect of patch attacks on the decision-making process of object detection models.
One Aspect of Problem-Solving Approach
[0035] The abnormality detection function according to the present embodiment therefore implements measurement of the inconsistency between an intermediate stage and a final stage of a pipeline that implements a machine learning task of object detection, from an aspect of quantitatively interpreting the effect of any type of abnormality on the decision-making process of object detection models.
[0036] Here, the measurement of the inconsistency is based on the point of view that when an input scene contains an abnormality, the decision-making process of object detection models becomes unstable and the instability is reflected in the presence and absence of objects throughout the entire pipeline of the object detection model.
[0037] Based on this point of view, the "perception" of the intermediate stage of the pipeline of the object detection model and the "perception" of the final stage of the pipeline of the object detection

model are measured.
[0038] FIG. **2** is a schematic diagram illustrating one aspect of a problem-solving approach. As illustrated in FIG. **2**, as an example of the “perception” of the intermediate stage, a saliency map generated by explainable AI (XAI) technique may be acquired. This saliency map maps a score that quantifies the objectness of an input scene, for example, the likelihood that an object is present. On the other hand, as an example of “perception” of the final stage, an object detection result (prediction result) output by the object detection model, such as bounding box, may be acquired.
[0039] For example, a first value and a second value are derived from the saliency map, as an example of indexes used to measure the inconsistency between the intermediate stage and the final stage. For example, as an example of the “first value”, an index (complete localization: CL) can be calculated which is a localized value of the saliency map corresponding to the entire input scene. As an example of the “second value”, an index (background localization: BL) can be calculated which is a localized value of the saliency map corresponding to the outside of a prediction result, for example, a bounding box, output by the object detection model in the input scene.
[0040] These first and second values can be used to quantify the effect of any type of abnormality on the decision-making process of object detection models.
[0041] Here, in FIG. **2**, as an example of an index (distinctive localization: DiL) that quantifies the effect of any type of abnormality on the decision-making process of object detection models, a third value, for example, the ratio of the first value and the second value is given. Furthermore, FIG. **2** illustrates an example of an deliberate abnormality type case in which a disappearance attack is executed by an adversarial patch that place on an object in an input scene to cause false recognition that this real object in the input scene did not exist.
[0042] In one aspect, when the disappearance attack works as intended by the attacker, the object disappears due to the adversarial patch. In this case, in the saliency map, the objectness score mapped to the saliency map is decreased by the amount of the object erased by the adversarial patch. The effect of this decrease in the objectness score appears as a decrease in the first value. This decrease in the first value corresponding to the denominator that defines the third value increases the third value.
[0043] In another aspect, when the disappearance attack makes the saliency map unstable, the adversarial patch attempts to lower the objectness score in a region corresponding to the position of the adversarial patch. As a result, the objectness score in a region other than the region of the adversarial patch in the input scene may accidentally increase. In this case, the first value does not necessarily decrease. On the other hand, even if the adversarial patch succeeds in false recognition of an object, the adversarial patch is unable to decrease the objectness score of the object that it attempts to erase, so the objectness score remains. The effect of the thus remaining objectness score appears as an increase in the second value. This increase in the second value corresponding to the numerator that defines the third value increases the third value.
[0044] In this way, the third value increases both when the disappearance attack works as intended by the attacker and when the disappearance attack makes the saliency map unstable. Thus, the effect of the abnormality caused by the disappearance attack on the decision-making process of object detection models can be quantitatively interpreted.
[0045] Although FIG. **1** illustrates the adversarial patch in the disappearance attack by way of example, the effect of the abnormality caused by partial occlusions or OODs. In addition, the effect of not only the abnormality deliberately caused by an adversarial patch but also the abnormality occurring naturally on the decision-making process of object detection models can be quantitatively interpreted.
[0046] The abnormality detection function according to the present embodiment therefore can detect any type of abnormality.
Configuration of Server Device **10**
[0047] A functional configuration example of the server device **10** according to the present

embodiment will now be described. FIG. **3** is a block diagram illustrating a functional configuration example of the server device **10**. FIG. **3** schematically depicts the blocks related to the functions corresponding to the automatic checkout service of the server device **10**.

[0048] As illustrated in FIG. **3**, the server device **10** includes a communication control unit **11**, a storage unit **13**, and a control unit **15**. FIG. **3** only depicts the functional units related to the functions corresponding to the automatic checkout service described above, and the server device **10** may include functional units other than those depicted in the drawing.

[0049] The communication control unit **11** is a functional unit that controls communication with other devices such as the camera **20** and the self-checkout register **30**. For example, the communication control unit **11** may be implemented by a network interface card. In one aspect, the communication control unit **11** accepts video data from the camera **20**. In another aspect, the communication control unit **11** accepts various requests related to the automatic checkout service from the self-checkout register **30** and outputs the processing results by the control unit **15** for various requests to the self-checkout register **30**.

[0050] The storage unit **13** is a functional unit that stores various data. For example, the storage unit **13** may be implemented by an internal, external or auxiliary storage of the server device **10**. For example, the storage unit **13** stores information such as an object detection model M.

[0051] The object detection model M may be a machine learning model that implements an object detection task that receives an image as an input and outputs the position and category of objects in the image. For example, the machine learning model may be implemented by you only look once (YOLO), or faster regions with convolutional neural network (Faster-RCNN).

[0052] To train such a machine learning model, a training dataset of training data can be used, in which an image is associated with a correct label that includes a bounding box indicating the position of an object in the image and a class indicating the category of the object.

[0053] In one aspect, in a training phase, the machine learning model can be trained according to a machine learning algorithm, for example, deep learning, using the image as an explanatory variable and the correct label as an objective variable. As a result, a trained machine learning model can be obtained.

[0054] In another aspect, in a prediction phase, an input scene is input into the trained machine learning model. The machine learning model receiving the input scene in this way outputs a bounding box, a class, and a confidence level.

[0055] Hereinafter, the trained machine learning model that implements the object detection task may be referred to as "object detection model" or "object detector (OD) model".

[0056] The control unit **15** is a functional unit that performs overall control of the server device **10**. For example, the control unit **15** may be implemented by a hardware processor. As illustrated in FIG. **3**, the control unit **15** includes an acquisition unit **15**A, an object detection unit **15**B, a map generation unit **15**C, a calculation unit **15**D, an abnormality detection unit **15**E, and a checkout processing unit **15**F. The control unit **15** may be implemented by hardwired logic or the like.

[0057] The acquisition unit **15**A is a processing unit that acquires an input scene. According to an embodiment, the acquisition unit **15**A can acquire video data from the camera **20** in units of a frame. For example, the acquisition unit **15**A may output each of the image frames acquired from the camera **20**, as an input scene, to the object detection unit **15**B.

[0058] The object detection unit **15**B is a processing unit that detects an object from an input scene. According to an embodiment, the object detection unit **15**B inputs an input scene to the OD model M to acquire an object detection result (prediction result) output by the OD model M.

[0059] FIG. **4** is a schematic diagram illustrating an example of object detection. As illustrated in FIG. **4**, the object detection unit **15**B inputs an input scene **40** to the OD model M. The OD model M receiving the input scene **40** in this way outputs an object detection result **40**R that includes a bounding box, a class, and a confidence level for each object detected from the input scene **40**.

[0060] For example, in the example of the object detection result **40**R, a "class name" such as Milk and a “confidence level” of 0 to 1 are predicted for each of bounding boxes BB**11** to B**13** detected from the input scene **40**.

[0061] The map generation unit **15**C is a processing unit that generates a saliency map corresponding to an input scene using the XAI technique. According to an embodiment, the XAI technique can use any saliency map output technique that is compatible with any computer-vision task. Examples include local interpretable model-agnostic explanations (LIME), Shapley additive explanations (SHAP), gradient-weighted class activation mapping (GradCAM), GradCAM++, and FACE.

[0062] Here, in an aspect, the method of calculating an objectness score varies according to the type of object detection algorithm used in the OD model M. Based on this aspect, the map generation unit **15**C selects an internal component Mo corresponding to the intermediate stage used in the measurement of the inconsistency in the pipeline of the OD model M, according to the type of object detection algorithm in the OD model M.

[0063] FIG. **5** is a schematic diagram illustrating an example of saliency map generation. As an example, FIG. **5** schematically illustrates how a saliency map **40**SE corresponding to the input scene **40** is generated by the XAI technique.

[0064] As illustrated in FIG. **5**, the type of object detection algorithm in the OD model M can be classified into a one-stage type, which performs region search and class classification simultaneously, and a two-stage type and a multi-stage type, in which a region proposal network is separated.

[0065] The one-stage type includes YOLO. In a one-stage type object detection algorithm, an objectness score is computed for each of bounding boxes that can overlap each other, along with the object's class and confidence level. Then, a function called non-maximum suppression (NMS) suppresses the output of a bounding box whose objectness score is not the largest, of the bounding boxes that overlap each other. In such a one-stage type OD model M, the final layer to which NMS has been applied is selected as the internal component Mo.

[0066] The two-stage type and multi-stage type include Faster-RCNN and cascade region proposal network (Cascade-RPN). In these two-stage type and multi-stage type OD models M, an objectness score is computed by an RPN component. Hence, the final layer of the RPN component is selected as the internal component Mo.

[0067] After the internal component Mo is selected in this way, the map generation unit **15**C computes the saliency map for the internal component Mo using a saliency map output technique S that has been user-defined or system defined.

[0068] In one aspect, when an activation-based saliency map output technique S is used, the map generation unit **15**C inputs, to the saliency map output technique S, an activation MA (x) in processing an input scene x from the first layer of the OD model M to the internal component Mo.

[0069] In another aspect, when a gradient-based saliency map output technique S is used, the map generation unit **15**C inputs, to the saliency map output technique S, a gradient M∇(x) in processing an input scene x from the first layer of the OD model M to the internal component Mo.

[0070] As an example, this gradient M∇(x) can be computed by back propagating an aggregation of objectness scores in the internal component Mo from the internal component Mo to the input scene x. In the back propagation, the computation corresponding to the type of object detection algorithm in the OD model M can be applied.

[0071] In the case of the one-stage type OD model M, the objectness scores generated by the internal component Mo are summed to a single scalar value. On the other hand, in the case of the two-stage type and multi-stage type OD models M, the outputs Mo(x) of the internal component when the input scene x is input to the OD model M are multiple vectors. Each of the vectors corresponds to a region of interest containing a plurality of bounding box candidates. To aggregate the outputs Mo(x) of the internal component into a single scalar value, the bounding box with the highest objectness score among the bounding box candidates is selected, and the objectness scores of the regions are summed. Thus, no region of interest is overlooked in the computation of the gradient M∇(x), resulting in gradients that are more indicative for the saliency map output technique S.

[0072] In this way, the activation MA (x) or the gradient M∇(x) is input to the saliency map output technique S, resulting in a saliency map of the internal component Mo. The saliency map of the internal component Mo is derived as a vector SM(x) that has the same dimensions as the number of pixels in the input scene x and reflects the objectness score of each pixel. For example, in the example illustrated in FIG. **5**, a saliency map **40**SM is generated. In the saliency map **40**SM, darker hatching is assigned to higher objectness scores of the pixels.

[0073] The calculation unit **15**D is a processing unit that calculates the first value and the second value that are used for measuring the inconsistency between the intermediate stage and the final stage of the OD model M. Hereinafter, a CL value will be given as an example of the first value and a BL value as an example of the second value.

[0074] In one aspect, the calculation unit **15**D calculates the CL value that is a localized value of the saliency map SM(x) corresponding to the entire input scene x. Such a CL value can be calculated according to the following equation (1) as an example. For example, "s.sub.i" in the following equation (1) refers to the i.sup.th value in the saliency map SM(x). According to equation (1), the CL value normalized to the numerical range of 0 to 1 can be derived. For example, as the CL value decreases, the input scene x is more likely to be an adversarial example, whereas as the CL value increases, the input scene x is more likely to be a benign example.

[00001] $$\text{CompleteLocalization} = \frac{{}_{s_i \in \text{SM}(x)} s_i}{\text{.Math. SM}(x)\text{ .Math.}} \quad (1)$$

[0075] In another aspect, the calculation unit **15**D calculates the BL value that is a localized value of the saliency map SM(x) corresponding to the outside of the bounding box output by the OD model M in the input scene x. Such a BL value can be calculated according to the following equation (2) as an example. For example, "s.sub.i" in the following equation (2) refers to the i.sup.th value in the saliency map SM(x). Furthermore, "BGS" in the following equation (2) refers to a set of pixels located outside the bounding box output by the OD model M, that is, pixels corresponding to the background of the object. According to equation (2), the BL value normalized to the numerical range of 0 to 1 can be derived. For example, as the BL value increases, the input scene x is more likely to be an adversarial example, whereas as the BL value decreases, the input scene x is more likely to be a benign example.

[00002] $$\text{BackgroundLocalization} = \frac{{}_{s_i \in \text{SM}(x)} \begin{matrix} s_i & \text{if} s_i \in \text{BGS} \\ 0 & \text{otherwise} \end{matrix}}{\text{.Math. SM}(x)\text{ .Math.}} \quad (2)$$

[0076] After these CL and BL values are calculated, the calculation unit **15**D calculates a DiL score based on the CL and BL values. Such a DiL score can be calculated according to the following equation (3), which finds a deviation of the CL and BL values, as an example. According to equation (3), the DiL score normalized to the numerical range of 0 to 1 can be derived. For example, as the DiL score increases, the input scene x is more likely to be an adversarial example, whereas as the DiL score decreases, the input scene x is more likely to be a benign example.

[00003] $$\text{DiL} = \frac{\text{BackgroundLocalization}}{\text{CompleteLocalization}} = \frac{{}_{s_i \in \text{SM}(x)} \begin{matrix} s_i & \text{if} s_i \in \text{BS} \\ 0 & \text{otherwise} \end{matrix}}{{}_{s_i \in \text{SM}(x)} s_i} \quad (3)$$

[0077] FIG. **6** to FIG. **9** are schematic diagrams (**1**) to (**4**) illustrating examples of DiL score calculation. In FIGS. **6** to **9**, FIG. **6** illustrates a benign example **40** as an example of the input scene x, while FIGS. **7** to **9** illustrate adversarial examples **41** to **43** containing adversarial patches Ad**1** to Ad**3** that execute a disappearance attack.

[0078] As illustrated in FIGS. **6** to **9**, the CL value for each example **40** to **43** can be calculated by summing and normalizing the values of all pixels in the saliency map **40**SM to **43**SM.

[0079] In the case of the benign example **40**, the CL value for the benign example **40** is calculated to be “0.67” by summing the objectness scores mapped corresponding to each of three objects as illustrated in the saliency map **40**SM in FIG. **6**.

[0080] In the case of the adversarial example **41**, one of three objects is erased by the adversarial patch Ad**1** as illustrated in the saliency map **41**SM in FIG. **7**. Thus, the CL value for the adversarial example **41** is calculated to be “0.41” by summing the objectness scores mapped corresponding to each of the remaining two objects. When the object disappears due to the adversarial patch Ad**1** in this way, the CL value of “0.41” for the adversarial example **41** is lower than the CL value of “0.67” for the benign example **40**. This clearly indicates that the effect of the disappearance attack by the adversarial patch Ad**1** on the decision-making process of the OD model M can be reflected as a decrease in the CL value.

[0081] In the case of the adversarial example **42**, as illustrated in the saliency map **42**SM in FIG. **8**, the adversarial patch Ad**2** attempts to lower the objectness score of the region corresponding to the position of the adversarial patch Ad**2**, resulting in an accidental increase in objectness score over a wide range other than the region of the adversarial patch Ad**2**. The CL value for the adversarial example **42** is calculated to be “0.9” by summing the objectness scores mapped in a wider range.

[0082] In the case of the adversarial example **43**, a complete disappearance attack is accomplished by the adversarial patch Ad**3**, as illustrated in the saliency map **43**SM in FIG. **9**. As a result, the objectness score for the entire scene is decreased. As a result, the CL value for the adversarial example **43** is calculated to be “0.01”.

[0083] On the other hand, the BL value for each example **40** to **41** can be calculated by summing and normalizing the values of pixels outside the region of the bounding box, that is, the black solid section in the saliency maps **40**SM to **43**SM, as illustrated in FIGS. **6** to **9**.

[0084] In the case of the benign example **40**, the objectness scores masked by the bounding boxes BB**11** to BB**13** corresponding to three objects are summed, as illustrated in the saliency map **40**SM in FIG. **6**. As a result, the BL value for the benign example **40** is calculated to be “0.06”.

[0085] In the case of the adversarial example **41**, one of three objects is erased by the adversarial patch Ad**1** as illustrated in the saliency map **41**SM in FIG. **7**.

[0086] Furthermore, the objectness scores masked by the bounding boxes BB**12** and BB**13** corresponding to the remaining two objects are summed. As a result, the BL value for the adversarial example **41** is calculated to be “0.09”.

[0087] In the case of the adversarial example **42**, as illustrated in the saliency map **42**SM in FIG. **8**, the adversarial patch Ad**2** attempts to lower the objectness score of the region corresponding to the position of the adversarial patch Ad**2**, resulting in an accidental increase in objectness score over a wide range other than the region of the adversarial patch Ad**2**. The mapped objectness scores thus increased in a wider range are partially masked by the bounding boxes BB**12** and BB**13** corresponds to two objects. However, the objectness score mapped corresponding to the object that the adversarial patch Ad**2** attempts to erase has a portion left without being lowered outside the region of the adversarial patch Ad**2**. As a result, the BL value for the adversarial example **42** is calculated to be “0.37”. In this way, the BL value of “0.37” for the adversarial example **42** is greater than the BL value of “0.06” for the benign example **40**. This clearly indicates that the behavior of the decision-making process of the OD model M that becomes unstable in quality of the saliency map **42**SM due to the adversarial patch Ad**2** can be reflected as an increase in the BL value.

[0088] In the case of the adversarial example **43**, a complete disappearance attack is accomplished by the adversarial patch Ad**3**, as illustrated in the saliency map **43**SM in FIG. **9**. As a result, the objectness score for the entire scene is decreased and no bounding box is obtained. As a result, the BL value for the adversarial example **43** is calculated to be “0.01”.

[0089] As illustrated in FIGS. **6** to **9**, the DiL scores for the examples **40** to **41** can be calculated using the thus calculated CL and BL values. In the DiL scores for the examples **40** to **41**, the DiL score “0.08” for the benign example **40** can be calculated as the minimum score. Furthermore, the values “0.21”, “0.41” and “1” higher than the DiL score “0.08” for the benign example **40** can be calculated as DiL scores for the adversarial examples **41** to **43**. This clearly indicates that it is possible to discriminate between the benign example **40** and the adversarial examples **41** to **43**.

[0090] Returning to the description of FIG. **3**, the abnormality detection unit **15**E is a processing unit that detects an abnormality contained in the input scene. According to an embodiment, the abnormality detection unit **15**E determines whether the DiL score calculated by the calculation unit **15**D exceeds a threshold Th, for example, 0.15. If the DiL score exceeds the threshold Th, the abnormality detection unit **15**E determines that there is an abnormality in the input scene x, that is, the input scene x is an adversarial example. On the other hand, if the DiL score does not exceed the threshold Th, the abnormality detection unit **15**E determines that there is no abnormality in the input scene x, that is, the input scene x is a benign example.

[0091] Here, when an abnormality is detected from the input scene x, the abnormality detection unit **15**E can perform a process corresponding to the abnormality. In one aspect, the abnormality detection unit **15**E can cause the self-checkout register **30** to output an alert. Examples of the alert include display or audio output of a message to warn of occurrence of partial occlusion or a message to prompt a retry to take a picture of the product. The output destination of the alert is not limited to the self-checkout register **30** and can be a terminal device (not illustrated) used by store personnel. In this case, the alert can be a display or audio output of a message to warn of occurrence of an attack using an adversarial patch, occurrence of partial occlusion, or detection of an object that does not fall into products of the store.

[0092] Such a threshold Th can be set based on a training dataset of the OD model M, as an example. For example, the DiL score is calculated from an image of training data, for each piece of training data contained in a training dataset of the OD model M. Then, the threshold Th can be set based on the statistics, for example, the maximum value of the DiL scores of all training data. For example, the maximum DiL score itself or a value obtained by adding a margin a, for example, 5% or 10% of the maximum DiL score, to the maximum DiL score can be set as the threshold Th.

[0093] The checkout processing unit **15**F is a processing unit that executes a process related to the automatic checkout service. In one aspect, if no abnormality is detected by the abnormality detection unit **15**E, the checkout processing unit **15**F performs image recognition of a product corresponding to an object contained in the input scene to update a shopping list displayed on the self-checkout register **30**.

Process Flow

[0094] FIG. **10** is a flowchart illustrating an abnormality detection process. This process can be initiated when the input scene x is acquired. As illustrated in FIG. **10**, the acquisition unit **15**A acquires the input scene x (step S**101**).

[0095] Subsequently, the object detection unit **15**B inputs the input scene x acquired at step S**101** to the OD model M (step S**102**). Thus, the object detection result (prediction result) output by the OD model M is acquired.

[0096] Subsequently, the map generation unit **15**C generates the saliency map SM(x) corresponding to the input scene acquired at step S**101**, using the XAI technique (step S**103**).

[0097] The calculation unit **15**D then calculates the CL value that is a localized value of the saliency map SM(x) corresponding to the entire input scene x (step S**104**A).

[0098] Furthermore, the calculation unit **15**D calculates the BL value that is a localized value of the saliency map SM(x) corresponding to the outside of the bounding box output by the OD model M in the input scene x (step S**104**B).

[0099] The processes at steps S**104**A and S**104**B may be executed in any order or may be executed concurrently.

[0100] The calculation unit **15**D then calculates the DiL score based on the CL value calculated at step S**104**A and the BL value calculated at step S**104**B (step S**105**).

[0101] The abnormality detection unit **15**E then determines whether the DiL score calculated at step S**105** exceeds the threshold Th, for example, 0.15 (step S**106**).
[0102] Here, if the DiL score does not exceed the threshold Th (Yes at step S**106**), it is determined that there is no abnormality in the input scene x, that is, the input scene x is a benign example. In this case, the checkout processing unit **15**F performs image recognition of a product corresponding to an object contained in the input scene x to update a shopping list displayed on the self-checkout register **30** (step S**107**) and terminates the process.
[0103] On the other hand, if the DiL score exceeds the threshold Th (No at step S**106**), the abnormality detection unit **15**E determines that there is an abnormality in the input scene x. In this case, the abnormality detection unit **15**E outputs an alert to any output destination such as the self-checkout register **30** (step S**108**) and terminates the process.
One Aspect of Effects
[0104] As described above, the server device **10** according to the present embodiment detects an abnormality in the input scene x, based on the CL value that is a localized value of the saliency map SM(x) corresponding to the entire input scene x and the BL value that is a localized value of the saliency map SM(x) corresponding to the outside of the bounding box output by the OD model M. Thus, the effect of any type of abnormality on the decision-making process of the OD model M can be quantitatively interpreted. The server device **10** according to the present embodiment therefore can detect any type of abnormality.
Second Embodiment
[0105] An embodiment of the present invention has been described above. The present invention may be implemented in various modifications in addition to the embodiment described above.
Numerical Values
[0106] The training data, input scene, objects, value of confidence level, and contents and number of image processing used in the above embodiment are only examples and can be changed as desired. The process described in each flowchart can also be modified as appropriate in a consistent manner. Models generated by various algorithms such as neural networks can be employed as the model.
System
[0107] The process procedures, control procedures, specific names, and information including various data and parameters referred to in the description and drawings may be changed as desired unless otherwise specified.
[0108] The specific forms of distribution and integration of the components of each device are not limited to those depicted in the drawings. For example, the calculation unit **15**D may be distributed into units that individually calculate the first, second, and third values. In other words, all or some of the components may be functionally or physically distributed or integrated in arbitrary units, depending on various loads and use conditions. Furthermore, the processing functions of each device can be entirely or partially implemented by a CPU and a computer program analyzed and executed by the CPU, or by hardware using wired logic.
Hardware
[0109] A hardware configuration example of a computer described in the foregoing first embodiment will now be described. FIG. **11** is a diagram illustrating a hardware configuration example. As illustrated in FIG. **11**, an information processing device **10** includes a communication device **10***a*, a storage device **10***b*, a memory **10***c*, and a processor **100***d*. The parts illustrated in FIG. **11** may be connected to each other by buses or other means.
[0110] The communication device **100***a* is a network interface card or the like. The storage device **10***b* is a storage device such as a hard disk drive (HDD) or a solid state drive (SSD). For example, the storage device **10***b* stores computer programs and DBs for implementing the functions illustrated in FIG. **3**.
[0111] A processor **10***d* operates a process that executes the functions illustrated in FIG. **3** by

reading a computer program for executing the same processing as the processing unit illustrated in FIG. **3** from the storage device **10***b* or the like and loading the computer program into the memory **10***c*.
[0112] Such a process implements the same functions as the processing units of the server device **10**. For example, the processor **10***d* reads, from the storage device **10***b* or the like, a computer program having the same functions as the acquisition unit **15**A, the object detection unit **15**B, the map generation unit **15**C, the calculation unit **15**D, the abnormality detection unit **15**E, the checkout processing unit **15**F, and the like. The processor **10***d* then executes a process that executes the same processing as the acquisition unit **15**A, the object detection unit **15**B, the map generation unit **15**C, the calculation unit **15**D, the abnormality detection unit **15**E, the checkout processing unit **15**F, and the like.
[0113] In this way, the information processing device **10** operates as an information processing device that executes a calculation method by reading and executing a computer program. Alternatively, the information processing device **10** can read the computer program from a recording medium by a medium reader and execute the read computer program to implement the same functions as the above embodiments. A computer program referred to in other embodiments is not necessarily executed by the information processing device **10**. For example, the process illustrated in FIG. **10** can be similarly applied when other computers or servers execute a computer program or when they cooperate to execute a computer program.
[0114] The computer program can be distributed via a network such as the Internet. Alternatively, the computer program can be recorded on any recording medium, read from the recording medium, and executed by a computer. For example, the recording medium can be implemented by a hard disk, a flexible disk (FD), a CD-ROM, a magneto-optical disk (MO), a digital versatile disc (DVD), or the like.
REFERENCE SIGNS LIST
[0115] According to an aspect of an embodiment, it is possible to detect any type of abnormality.
[0116] All examples and conditional language recited herein are intended for pedagogical purposes of aiding the reader in understanding the invention and the concepts contributed by the inventors to further the art, and are not to be construed as limitations to such specifically recited examples and conditions, nor does the organization of such examples in the specification relate to a showing of the superiority and inferiority of the invention. Although the embodiments of the present invention have been described in detail, it should be understood that the various changes, substitutions, and alterations could be made hereto without departing from the spirit and scope of the invention.

## Claims

**1**. A non-transitory computer-readable recording medium having stored therein an abnormality detection program that causes a computer to execute a process comprising: calculating a first value and a second value, based on a detection result of an object detected from an input scene to an object detection model and a saliency map for the input scene obtained by an XAI technique, the first value indicating a value of the saliency map for the entire input scene, the second value indicating a value of the saliency map in a region other than a region of the detected object in the input scene; and detecting an abnormality in the input scene based on the calculated first and second values.
**2**. The non-transitory computer-readable recording medium according to claim 1, wherein the calculating includes calculating a third value based on the first and second values, and the detecting includes detecting an abnormality in the input scene based on the third value.
**3**. The non-transitory computer-readable recording medium according to claim 2, wherein the third value is a deviation of the first and second values.
**4**. The non-transitory computer-readable recording medium according to claim 1, wherein the XAI

technique is an activation-based or gradient-based saliency map output technique.

**5**. The non-transitory computer-readable recording medium according to claim 1, wherein the process further includes generating the saliency map for a layer to which non-maximum suppression (NMS) has been applied, in a one-stage type of the object detection model.

**6**. The non-transitory computer-readable recording medium according to claim 1, wherein the process further includes generating the saliency map for a final layer of a region proposal network (RPN) component in a two-stage type or multi-stage type of the object detection model.

**7**. An abnormality detection method that causes a computer to execute a process comprising: calculating a first value and a second value, based on a detection result of an object detected from an input scene to an object detection model and a saliency map for the input scene obtained by an XAI technique, the first value indicating a value of the saliency map for the entire input scene, the second value indicating a value of the saliency map in a region other than a region of the detected object in the input scene; and detecting an abnormality in the input scene based on the calculated first and second values, by processing circuitry.

**8**. An information processing device comprising: a processor configured to: calculate a first value and a second value, based on a detection result of an object detected from an input scene to an object detection model and a saliency map for the input scene obtained by an XAI technique, the first value indicating a value of the saliency map for the entire input scene, the second value indicating a value of the saliency map in a region other than a region of the detected object in the input scene; and detect an abnormality in the input scene based on the calculated first and second values.